1 DDTA BA2 relation/action vocabulary - R1

Status: CLOSED BY BA2-T4 / ACCEPTED FOR CURRENT THESIS SCOPE

Closure baseline reviewed: `d16743a6417196ebf53840b1210a645e9dda4245`

Identity dependency: `BA1_MINIMAL_BAE_IDENTITY_ONTOLOGY_R1.md` (`BAReferent + BAProposition`)

Historical derivation: BA2-T1 structural lower bound -> BA2-T2 vocabulary pressure test -> BA2-T3 cross-corpus regression -> BA2-T4 closure review.

1.1 1. Closure statement

BA2 closes the minimum methodology-neutral proposition/relation/action semantics required by the current DDTA thesis evidence.

The closure does **not** claim to define a universal systems-modeling language or an exhaustive verb ontology. It closes a governed Base Analysis semantic contract sufficient for the reviewed facial-access and order-fulfillment corpora and for the bounded method-neutral analysis-consumer pressure already performed.

The accepted proposition shape is:

```
BAProposition
|- semanticOperatorKey   1
|- participation         1..*
|    |- roleKey          1
|    `- term             1
|- polarity              1
`- scopedModifier        0..*    [condition / temporalScope only]
```

`semanticOperatorKey`, `roleKey`, `polarity` and `scopedModifier` are semantic vocabulary/structure, not new BAE identity families.

A participation `term` is normally a `BAReferent`. A controlled typed local value is permitted only when the meaning does not require independent identity across propositions, projections or change. If it does, BA1 requires promotion to `BAReferent` identity.

1.2 2. Accepted current-scope semantic operator registry

The following thirteen operator concepts are accepted as the current-scope minimum:

```
transfer
produce
create
observe
transition
correlate
reference
dependOn
consumeService
realize
assignResponsibility
constrain
classify
```

Each key has a stable methodology-neutral meaning. Human-readable wording, authoring verbs and synonyms are not semantic identity and remain downstream lexical concerns.

1.2.1 2.1 Normative meanings

Operator	Accepted methodology-neutral meaning
`transfer`	Assert conveyance of content from a source to one or more destinations.
`produce`	Assert that an actor/capability makes one or more governed results or outputs available, optionally from explicit inputs.
`create`	Assert establishment of a new project-semantic item or event occurrence.
`observe`	Assert inspection/read/query of existing project state or meaning without asserting creation or state change by the observation itself.
`transition`	Assert governed state/lifecycle change of a subject.
`correlate`	Assert same-request/evaluation/operation/context identity matching required to bind items correctly.
`reference`	Assert an explicit reference from one project meaning to another without by itself implying correlation, dependency or ownership.
`dependOn`	Assert that one project meaning requires another meaning/result/capability/milestone as a prerequisite.
`consumeService`	Assert actual consumption/use of a capability or service without transferring its ownership or responsibility.
`realize`	Assert that a more concrete project meaning realizes/materializes an abstract project meaning.
`assignResponsibility`	Assert placement or denial of a governed responsibility/authority kind over a project-semantic scope.
`constrain`	Assert a reusable or independently queryable restriction on behavior, state, acceptance, applicability or other governed project meaning.
`classify`	Assert assignment of a reusable method-neutral semantic kind to a `BAReferent`.

1.3 3. Accepted operator-scoped role contracts

Role concepts may recur across operators, but role validity and cardinality are operator-scoped. A context-free global role contract is rejected.

Operator	Required roles	Optional roles	Accepted current-scope cardinality
`transfer`	`source`, `destination`, `content`	none	source `1`; destination `1..*`; content `1..*`
`produce`	`actor`, `result`	`input`	actor `1`; result `1..*`; input `0..*`
`create`	`actor`, `created`	none	actor `1`; created `1..*`
`observe`	`actor`, `observed`	`result`	actor `1`; observed `1..*`; result `0..*`
`transition`	`subject`, `toState`	`actor`, `fromState`	subject `1`; toState `1`; actor `0..1`; fromState `0..1`
`correlate`	`correlatedItem`, `correlationContext`	none	correlatedItem `1..*`; correlationContext `1`
`reference`	`referencer`, `referenced`	none	referencer `1`; referenced `1..*`
`dependOn`	`dependent`, `prerequisite`	none	dependent `1`; prerequisite `1..*`
`consumeService`	`consumer`, `service`	`provider`	consumer `1..*`; service `1`; provider `0..1`
`realize`	`abstract`, `realization`	none	abstract `1`; realization `1..*`
`assignResponsibility`	`responsibleParty`, `responsibilityScope`, `responsibilityKind`	none	responsibleParty `1..*`; responsibilityScope `1`; responsibilityKind `1`
`constrain`	`constraintTarget`, `constraintValue`	none	constraintTarget `1`; constraintValue `1..*`
`classify`	`classifiedReferent`, `semanticKind`	none	classifiedReferent `1`; semanticKind `1..*`

These cardinalities are accepted for the current thesis scope. They are not claims that all conceivable future projects must use these maxima. A future governed corpus may reopen the smallest affected BA2 contract if it demonstrates that widening or splitting one role contract is required to preserve method-neutral project meaning.

1.4 4. Responsibility and ownership boundary

`ownOrManage` is not a separate operator.

Ownership, management and other authority modes are represented through `assignResponsibility` plus a controlled `responsibilityKind` and explicit polarity.

Conceptually:

```
operator: assignResponsibility
responsibleParty    -> project
responsibilityScope -> underlying transport infrastructure
responsibilityKind  -> ownership / management
polarity            -> negative
```

This preserves the facial-access distinction that the project can consume transport while not owning or managing the underlying infrastructure:

```
consumeService != responsibility/ownership
```

1.5 5. Explicit polarity

Every proposition exposes normalized polarity:

```
polarity = affirmative | negative
```

Polarity prevents proliferation of `notX` operator keys and keeps negation separate from operator identity.

A negative proposition does not eliminate the need for `constrain`. Normative prohibitions whose content is itself a governed reusable rule remain explicit constraint propositions.

1.6 6. Scoped modifier lower bound

Only two embedded modifier kinds are accepted in the BA2 lower bound:

- `condition` - a proposition-local applicability/guard condition with no independent reuse or identity;
- `temporalScope` - a proposition-local before/after/during/until scope with no independent reuse or identity.

A modifier may remain embedded only when all of the following hold:

1. it changes only the interpretation/applicability of one proposition or participation binding;
2. it introduces no independent participant set;
3. it needs no independent assertion-level provenance/review/change identity;
4. it is not reused elsewhere as project meaning.

If any condition fails, the meaning is represented through `BAReferent` and/or a separate `BAProposition`.

The following are therefore **not** default embedded modifiers when governed or analytically relevant:

- completion/failure semantics;
- atomicity;
- concurrency;
- idempotency;
- reusable applicability rules;
- realization;
- correlation;
- responsibility/authority;
- service consumption;
- reusable security or other specialized constraints.

They are represented through explicit propositions, normally `constrain` or another accepted operator as appropriate.

1.7 7. Classification contract

Method-neutral referent classification is represented through a normal proposition:

```
operator: classify
classifiedReferent -> <BAReferent>
semanticKind       -> <controlled method-neutral semantic key/value>
```

BA2 closes the **registry contract**, not an exhaustive universal semantic-kind taxonomy.

A semantic-kind registry entry must have:

1. a stable semantic key;
2. a normative method-neutral definition;
3. enough distinction for a consumer not to reconstruct the kind from raw prose;
4. no method-specific meaning such as STRIDE category, DFD element or tool class.

Examples under current pressure include party/person, capability, information/resource, behavior/event, store and contract. These examples do not constitute a universal closed enumeration and do not become BA1 metaclasses.

Classification propositions may later receive provenance, status and change disposition in BA3 while the classified `BAReferent` identity remains stable.

1.8 8. Semantic key versus lexical wording

BA2 closes this separation:

```
source lexical wording
        !=
stable BA semantic key
        !=
display label / synonym
```

For example, `deliver`, `send`, `submit` and `emit` may map to `transfer` when they express conveyance. `reads` and `queries` may map to `observe`. `maps` need not create a `map` operator when the governed meaning is production of a normalized result from an input plus a constraint.

The document-to-semantic mapping belongs to BA3 derivation mechanics. Display labels and authoring synonyms belong to BA5. Neither changes the accepted BA2 semantic key merely because wording changes.

1.9 9. General logical composition

Current evidence does not require a general-purpose logical-expression language inside BA2.

Compound governed clauses may derive to multiple `BAProposition` identities sharing the relevant `BAReferent` meanings. N-ary participation preserves the scope of each assertion; `condition`, `temporalScope`, polarity and explicit `constrain` propositions preserve the reviewed conditional, negative and normative rule pressure.

A future corpus must reopen BA2 if it demonstrates that a governed atomic assertion cannot be preserved without an explicit logical-composition construct and cannot be represented honestly by the accepted proposition shape plus separate reusable propositions.

1.10 10. Current-scope completeness and extension rule

The thirteen-key registry is **closed for current thesis scope**, but it is not a universal verb ontology.

A new operator key is justified only if concrete governed evidence demonstrates all of the following:

1. the project meaning is methodology-neutral;
2. existing operators plus roles, polarity, constraints and local modifiers cannot preserve the distinction without semantic loss or raw-prose reconstruction;
3. the distinction is needed for reusable human/method consumption or change reasoning;

4. the distinction is not merely lexical, implementation-specific, document-layer-specific or analysis-method-specific.

If those conditions are met, reopen only the affected BA2 registry/role contract. Do not broaden Base Analysis into a general systems-modeling language.

A role contract is reopened only when a concrete governed case cannot be represented faithfully under the accepted operator-scoped roles/cardinalities. A modifier kind is added only when it remains proposition-local under the modifier-admissibility rule and cannot be represented more honestly through an explicit proposition.

1.11 11. Relationship to BA1 and later phases

BA2 does not reopen BA1. `BAReferent` and `BAProposition` remain the only first-class semantic identity families.

BA2 does not define:

- source locators, baseline provenance or grounded/derived/diagnostic state - BA3;
- accepted/stale/superseded assertion lifecycle and equivalence mechanics - BA3;
- human or method-specific views/projections - BA4;
- lexical labels, synonyms or optional authoring/extraction assistance - BA5;
- complete holdout regression and final Base Analysis closure - BA6;
- STRIDE/STRIDE-AI categories, AnalysisRecord, Finding/Common Finding or ThreatForge classes - downstream analysis/tooling work.

Governed project documentation remains project authority. Base Analysis remains the accepted/rebuildable analytical representation of that governed meaning.

1.12 12. Camera example under the closed contract

A facial-access branch can expose, conceptually:

```
transfer
  source      -> CameraSubsystem
  destination -> RecognitionProcessor
  content     -> RecognitionCapture

correlate
  correlatedItem     -> RecognitionCapture
  correlationContext -> RecognitionRequest

consumeService
  consumer -> project endpoint context
  service  -> LocalConnectivity

assignResponsibility [negative]
  responsibleParty    -> project
  responsibilityScope -> underlying transport
  responsibilityKind  -> ownership/management

realize
  abstract    -> LocalConnectivity
  realization -> WiredEthernet

constrain
```

```
  constraintTarget -> delivery behavior
  constraintValue  -> incomplete delivery != successful completion

constrain
  constraintTarget -> delivery behavior
  constraintValue  -> confidentiality / integrity / authorized provenance
```

A later threat-method overlay may project these facts into its own constructs. The overlay does not redefine the shared Base Analysis semantics.

1.13 13. Closure disposition

```
BA2 proposition shape                         ACCEPTED
Stable methodology-neutral semantic key       ACCEPTED
Explicit role-bound n-ary participation        ACCEPTED
Current-scope 13-key operator registry         ACCEPTED
Operator-family facet in normative core        REJECTED
Operator-scoped role/cardinality contracts     ACCEPTED
Explicit polarity                              ACCEPTED
Embedded modifiers: condition/temporalScope    ACCEPTED
Generic/untyped modifier bag                   REJECTED
Governed reusable rule -> constrain             ACCEPTED
Classification-as-proposition                  ACCEPTED
Universal semantic-kind taxonomy               NOT REQUIRED
Semantic key != lexical wording                CLOSED
Third BA1 identity family                      NOT FORCED
BA1                                            REMAINS CLOSED
BA2                                            CLOSED BY BA2-T4
```